

- 2 In mammals, the small intestine is the main site of absorption of the products of digestion.

Fig. 2.1 is a transmission electron micrograph of a longitudinal section (L.S.) of part of an epithelial cell from the small intestine of a mammal.

Fig. 2.2 is a transmission electron micrograph of a horizontal section made at the position indicated by the two arrows in Fig. 2.1.

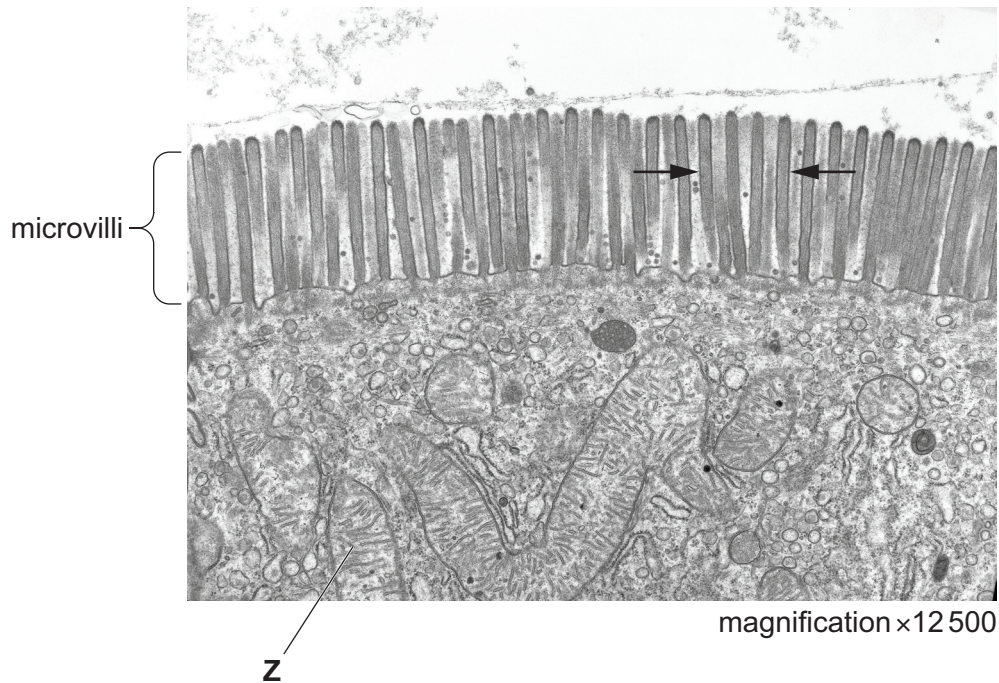


Fig. 2.1

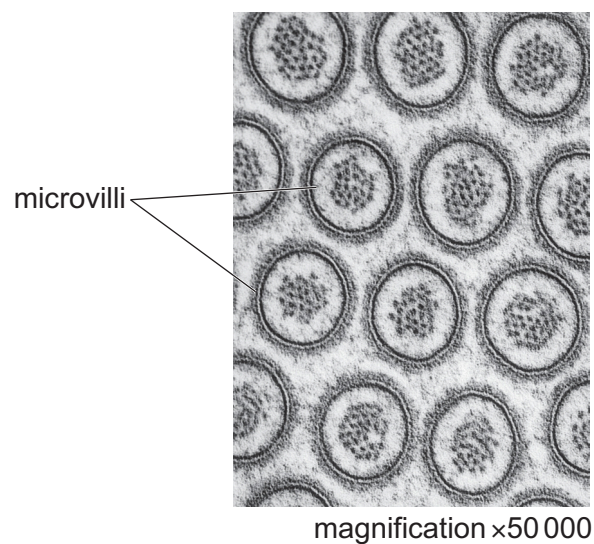


Fig. 2.2



- (a) Microvilli and cilia are cell structures.

Describe how the structure of cilia differs from the structure of the microvilli visible in Fig. 2.1 and Fig. 2.2.

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- (b) A scientist measured the length and the diameter of some of the microvilli shown in Fig. 2.1 to estimate the total surface area of microvilli on the surface of the epithelial cell.

The scientist assumed that each microvillus was cylindrical in shape.

Suggest **one** other measurement needed to estimate the total surface area of the microvilli of the epithelial cell.

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- (c) Identify the organelle labelled **Z** in Fig. 2.1 **and** explain why there is a large number of these organelles in the epithelial cells of the small intestine.

organelle

explanation

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- (d) Bacteria are found attached to epithelial cells in the intestines of mammals.

Describe how the organisation and distribution of DNA in epithelial cells differs from the organisation and distribution of DNA in bacterial cells.

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